# Introduction to Economics

Class 8 — Surplus, efficiency, taxes and subsidies

**Pavel Kocourek**

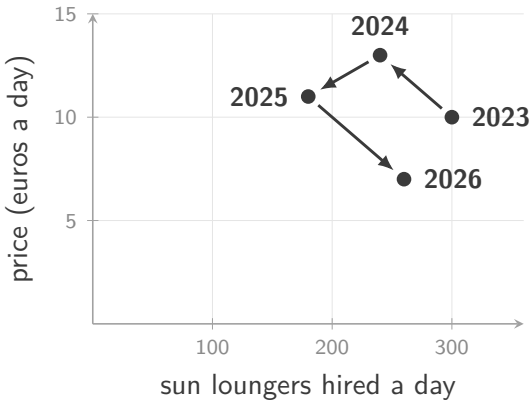
Universitat de Barcelona · Group A6

19 October 2026

# Plan for today

Warm-up; Class 6's coffee: just right?
Bread, milk, eggs: VAT down to zero
What buyers gain: consumer surplus
What sellers gain: producer surplus
In pairs: menús del día
Total surplus: efficient, but fair?
A tax: a wedge between two prices
Break
Who pays the tax office; where the money goes
In pairs: a tax on menús
Who bears a tax: elasticity decides; in pairs
A subsidy; VAT; the bread answered
In one line; before Thursday 22 October

# Warm-up: two dots — which curve moved?




## Sun loungers on one beach, each August

Each year only one curve moved. Which one, which way?

1. 2023 → 2024
2. 2024 → 2025
3. 2025 → 2026

In pairs, two minutes; then hands per arrow.

# Warm-up: what does playing cost you?

## Saturday: a volleyball tournament

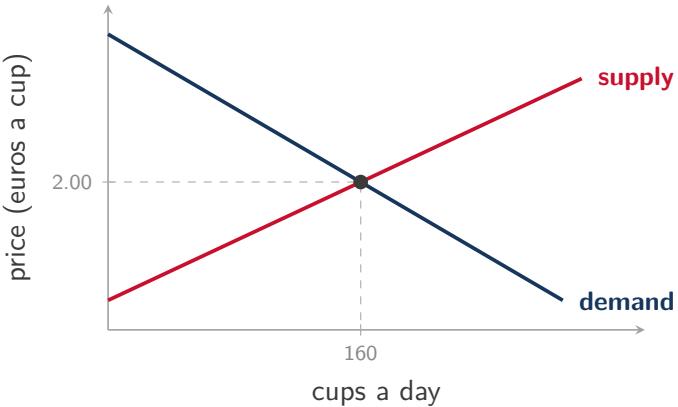
Entry, already paid: 12 euros, no refund
Same hours: an usher's shift at a film festival, 40 euros
Bus to the tournament and back: 4 euros

1. What does playing cost you?

2. Playing is worth 42 to you. Play?

In pairs, one minute; then hands.

# Class 6's coffee: 160 cups — too few, too many, or just right?



# 1 January 2023: VAT on bread, milk and eggs fell from 4 % to 0 %




bread, milk, cheese, eggs, fruit, vegetables

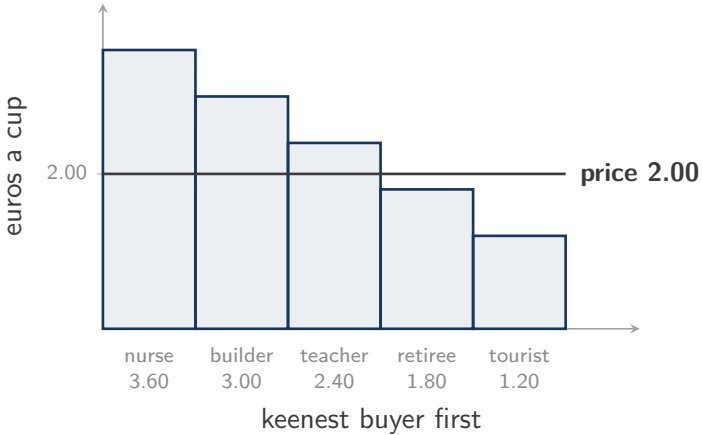



olive and seed oils, pasta

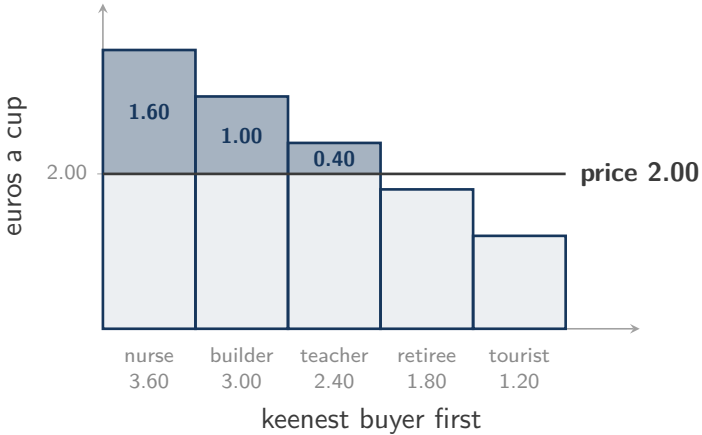
**A 1.04 € loaf: 4 cents of VAT.**
**How many reached the shopper?**

Real Decreto-ley 20/2022 (BOE, 28 December 2022), in force 1 January 2023.

# Willingness to pay: each step of demand is one buyer

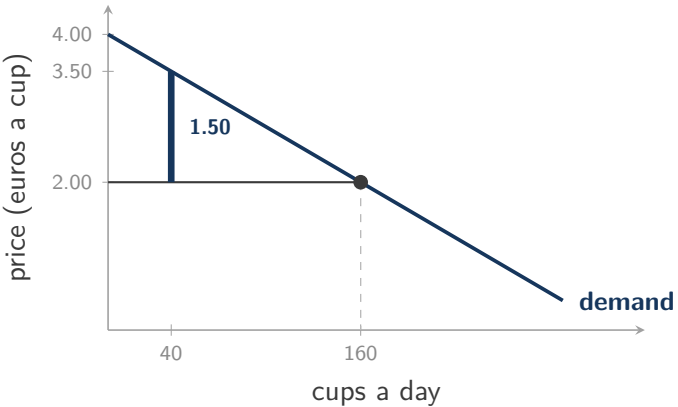


# Willingness to pay: each step of demand is one buyer

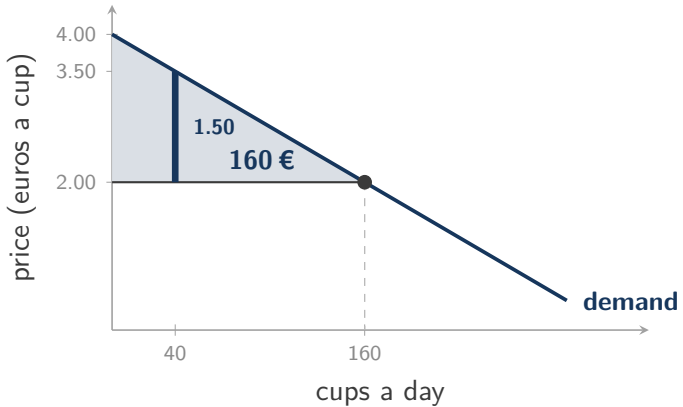



**Consumer surplus** = willingness to pay − price: $1.60 + 1.00 + 0.40 = 3.00\,€$.

# Consumer surplus: under demand, above the price — 160 € a day

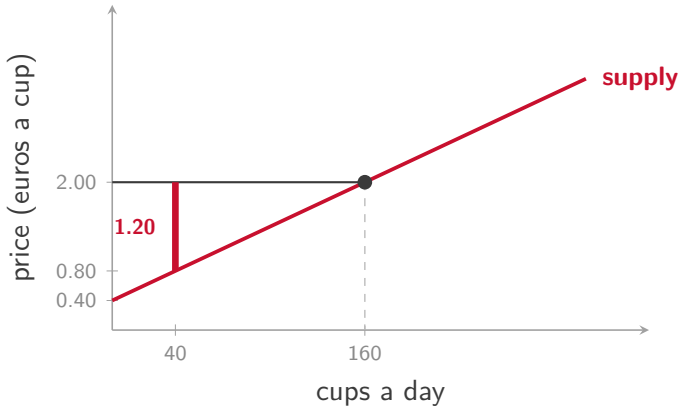


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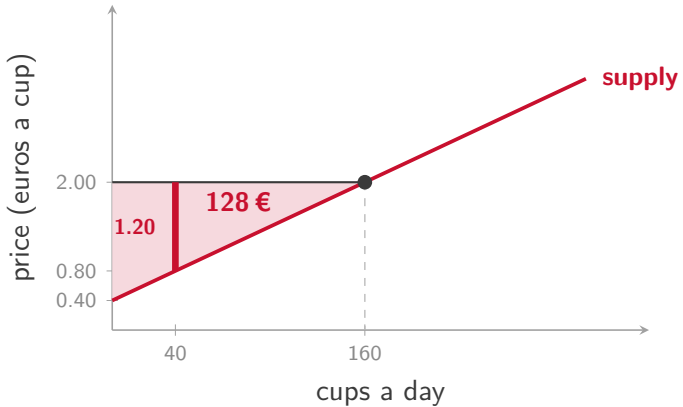



½ × 160 cups × 2.00 € = 160 euros a day: every buyer's bargain, added up.

# Producer surplus: above supply, below the price — 128 € a day

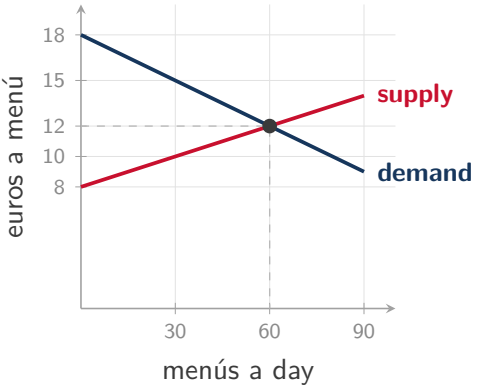


# Producer surplus: above supply, below the price — 128 € a day




½ × 160 cups × 1.60 € = 128 euros a day: what the bars get above their costs.

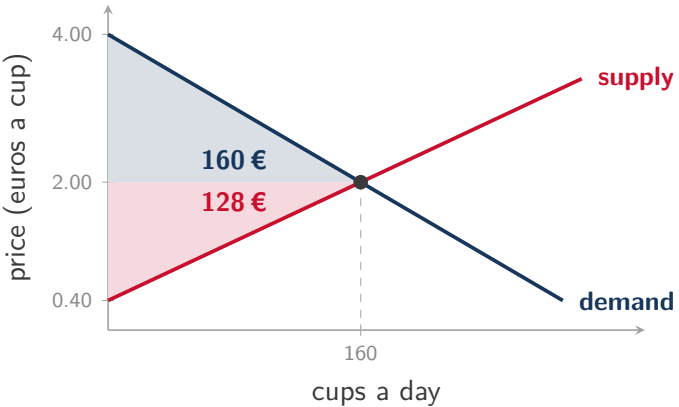
# In pairs, four minutes: read the surplus off the graph




## Menús del día on one street

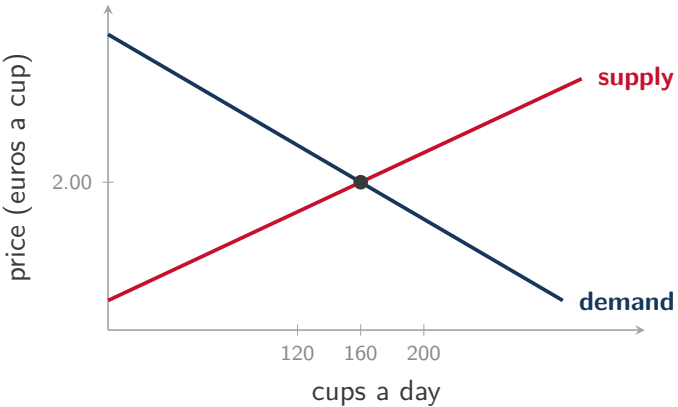
1. Consumer surplus at 12 €?
2. Producer surplus at 12 €?
3. The 30th menú: worth? costs?

# Total surplus: buyers' 160 plus the bars' 128 — 288 euros a day

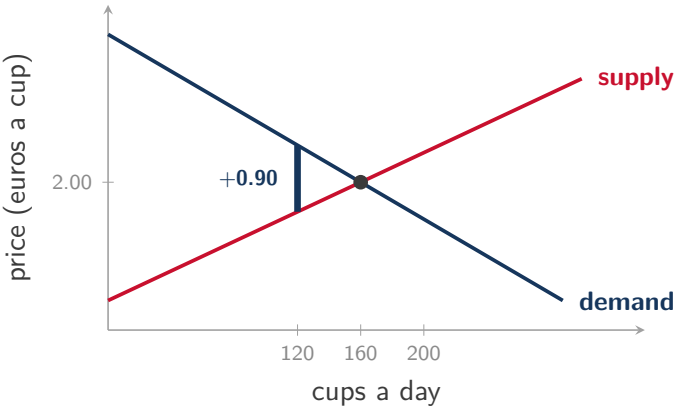



**Total surplus** = value to buyers − cost to sellers = 160 + 128 = 288 euros a day.

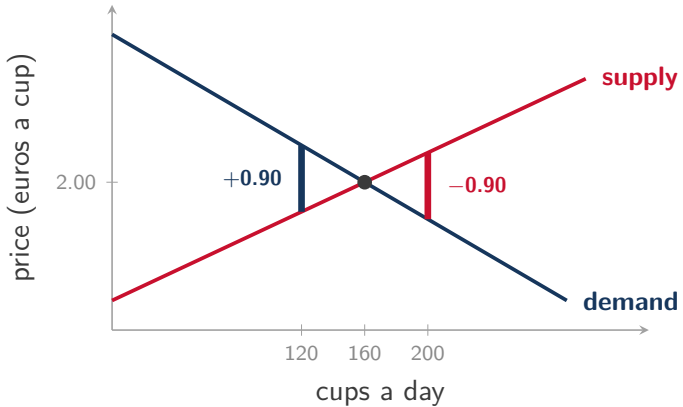
## Stop at 160: before it, cups worth more than they cost; after it, less



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Below 160: surplus left on the table. Above: cups that cost more than they are worth.

## Just right: the market's 160 cups give the largest total surplus

- The cups go to those who value them most
- They are made by the bars that make them cheapest
- Every cup worth more than its cost, and no other

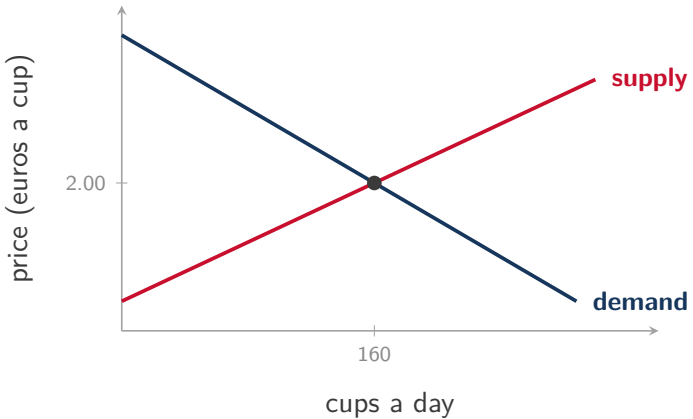
That is what economists call **efficient**, as long as no trade affects others (Class 10).

## Efficient is not the same as fair

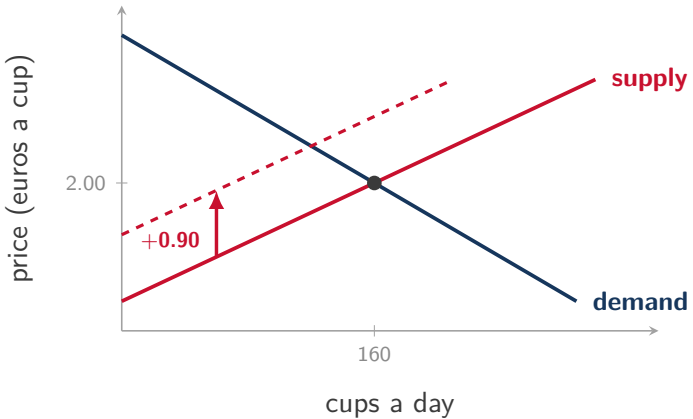
- Efficiency: is the cake as big as it can be?
- Equity: is it shared fairly?
- Willingness to pay depends on income

Efficiency is a positive question; fairness a normative one (Class 4).

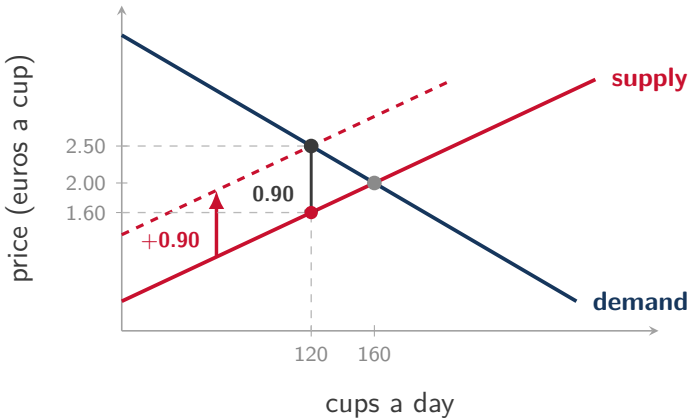
# A tax of 0.90 a cup: buyers pay 2.50, the bars keep 1.60



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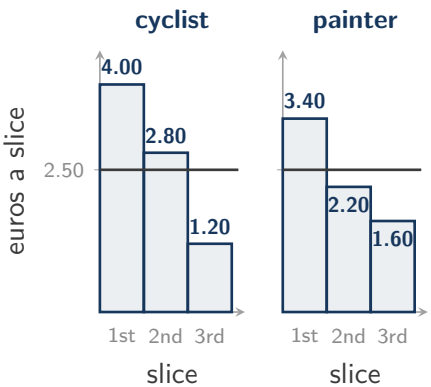


# A tax of 0.90 a cup: buyers pay 2.50, the bars keep 1.60




Buyers pay 0.50 more, the bars keep 0.40 less: a tax is a wedge between two prices.

# Market demand from two staircases (optional)



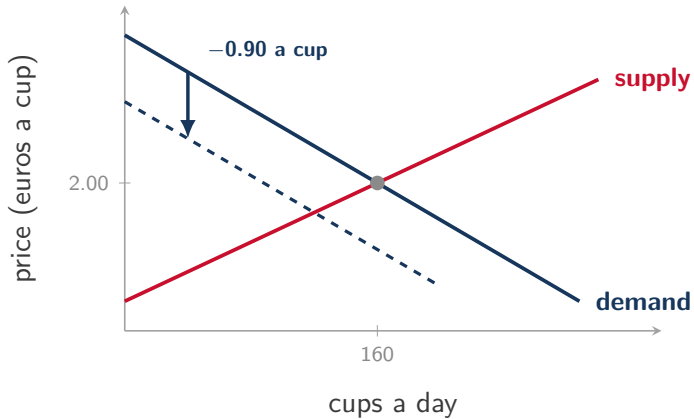

## Pizza slices: two buyers

1. At 2.50: slices bought in all?
2. Their consumer surplus at 2.50?
3. Market demand at 2.00?

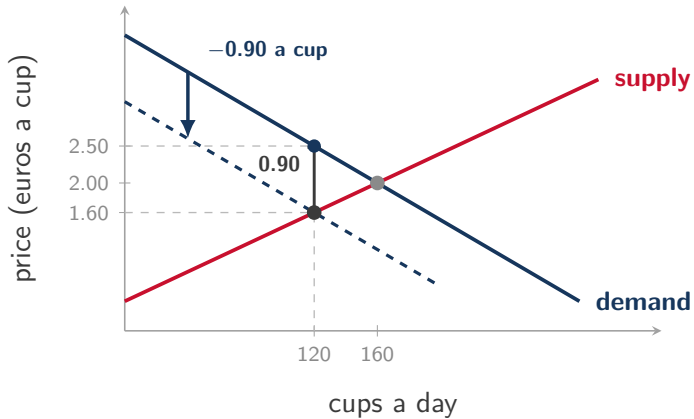
# Break

Ten minutes

# Tax the buyers instead: the same 120 cups, the same 2.50 and 1.60

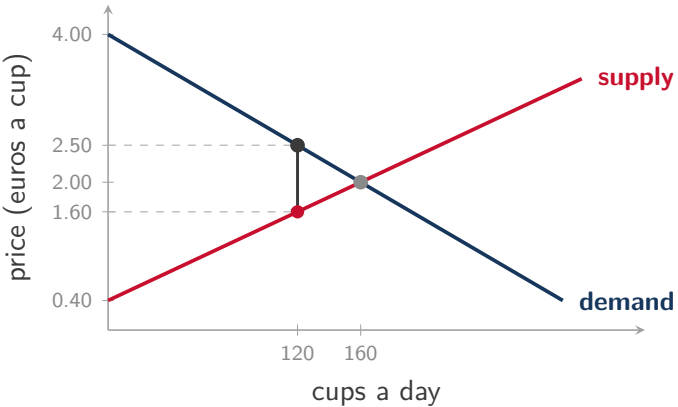


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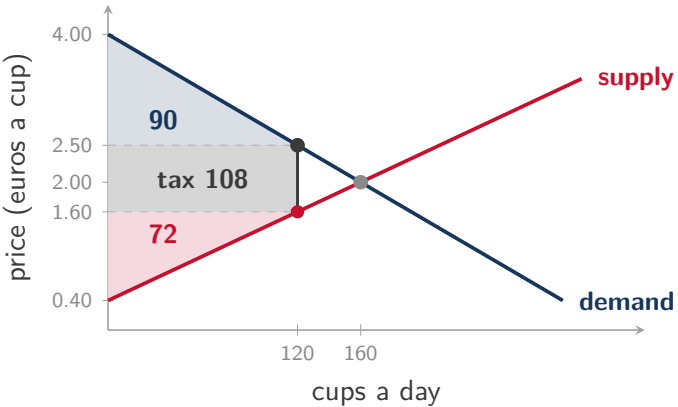



Who hands the 0.90 to the tax office does not matter: the wedge does.

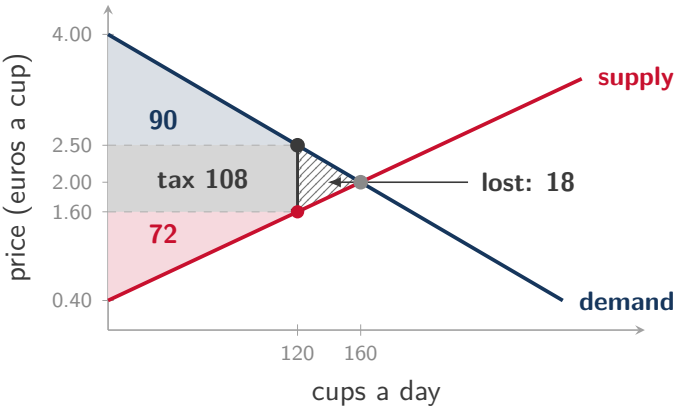
# With the tax: buyers 90, bars 72, the state 108 — and 18 lost



# With the tax: buyers 90, bars 72, the state 108 — and 18 lost



# With the tax: buyers 90, bars 72, the state 108 — and 18 lost




Buyers lose 70, the bars 56; the state gets 108. The lost 18: the **deadweight loss**, Thursday.

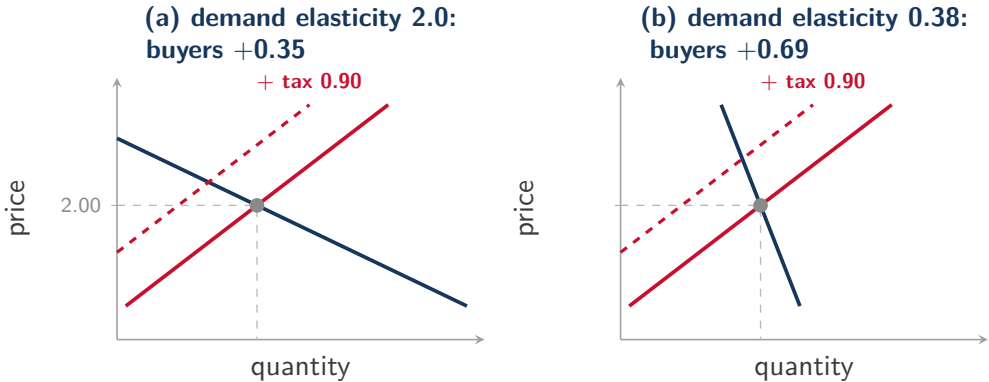
## In pairs, four minutes: a tax of 2.50 on every menú

### Same street, same menús

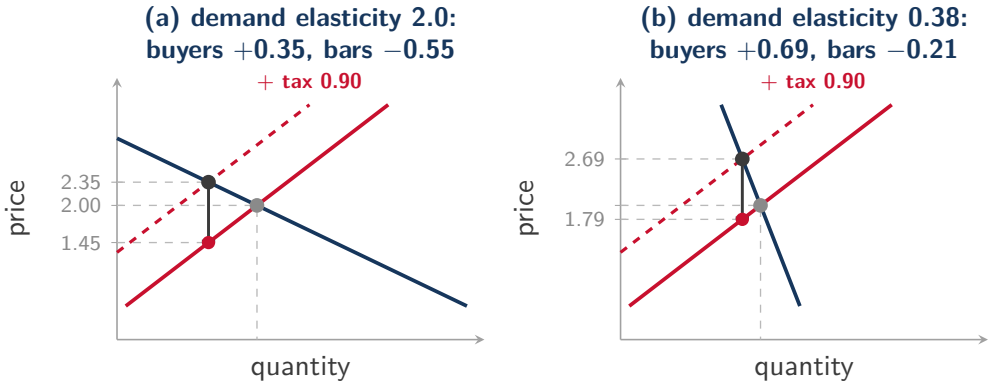
| menús a day | the last one is worth | the last one costs |
|---|---|---|
| 30 | 15.00 | 10.00 |
| 45 | 13.50 | 11.00 |
| 60 | 12.00 | 12.00 |

1. No tax: price and menús?
2. Tax 2.50: menús, and each side's price?
3. Tax collected a day?
4. Who bears more of the 2.50?

# Class 7's markets, taxed 0.90: who cannot walk away pays more



# Class 7's markets, taxed 0.90: who cannot walk away pays more




Buyers who can switch leave most of the tax to the bars; buyers who cannot pay most of it.

## Tax incidence: the less elastic side bears more of the tax

| | most of the tax falls on |
|---|---|
| **demand less elastic** | buyers |
| **supply less elastic** | sellers |
| **equally elastic** | half each |

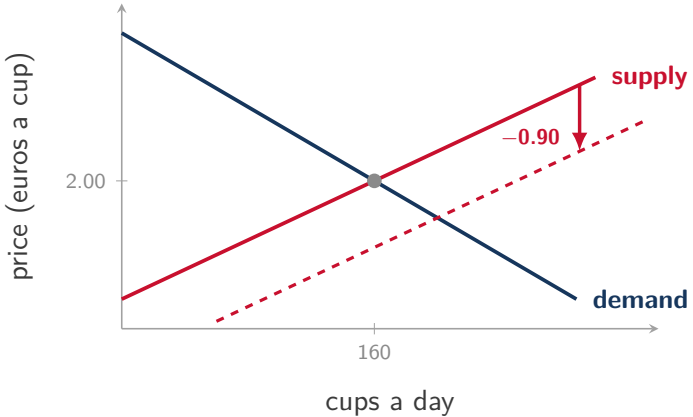
The law says who pays the tax office; elasticities say who bears the tax.

## In pairs, three minutes: who bears most of the tax?

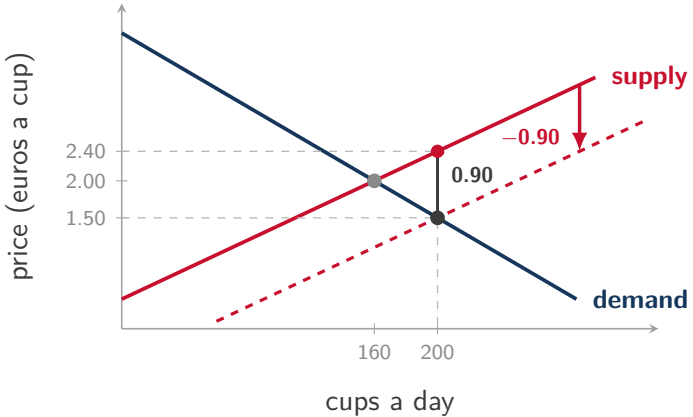
### Buyers or sellers — and why?

1. Cigarettes: demand elasticity 0.4, supply 2.0
2. Beach umbrellas on a July Sunday: demand 3.0, supply 0.5
3. Tax 1.00; buyers' price 5.00 $\rightarrow$ 5.80. Sellers keep? Less elastic side?
4. The same tax, paid by the buyers: price on the label?

# A subsidy of 0.90 a cup: buyers pay 1.50, the bars get 2.40



# A subsidy of 0.90 a cup: buyers pay 1.50, the bars get 2.40




The tax reversed: buyers save 0.50, the bars gain 0.40.
The state pays $0.90 \times 200 = 180\,€$ a day.

## Per unit or per cent: VAT is a percentage of the price

- A fixed sum per unit: a **specific tax** (0.90 a cup)
- A share of the price: **ad valorem** (VAT)
- Spain's VAT: 21 %, 10 %, 4 %
- A loaf: $1.00 + 4\,\%$ VAT $= 1.04\,€$

Either way a wedge; the elasticities decide who bears it.

# Revealed in class

Before class: by slide 20's rule, who should get most of a VAT cut on bread — shoppers or shops? Bread is a necessity, and a supermarket can stock more at about the same cost.

## Which curve moves? (optional)

### Cinema tickets in one city

1. A tax of 1.00 a ticket, paid by the cinemas: which curve shifts?
2. Tickets get dearer: does demand shift too?
3. No tax; a subsidy makes streaming cheaper. Ticket demand?
4. And the ticket price, and the tickets sold?

# Who gains from trade? (optional)

## Hours for one unit

| | 1 kg of cheese | 1 bottle of wine |
|---|---|---|
| **valley farm** | 2 | 1 |
| **hill farm** | 3 | 6 |

1. Who should make the cheese, and why?
2. At 1 bottle a kg: each farm's gain?
3. Hill farm at 4 h a kg, 2 h a bottle: any gain?

**Surplus: value minus price for buyers, price minus cost for sellers.**

**The market's quantity makes total surplus as large as it can be.**

A tax or a subsidy is a wedge; the less elastic side feels most of it.

Deadweight loss and price controls: Thursday.

## Before Thursday 22 October

- Thursday: price controls; international trade and tariffs
- Read Mankiw & Taylor: Ch. 7, pp. 159–163 and 172–178; Ch. 17, pp. 380–390
- Problem set 1: due Thursday 22 October, on paper, in class
- Practise in Sextant: the units the course page lists



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